

Midterm Examination Answer

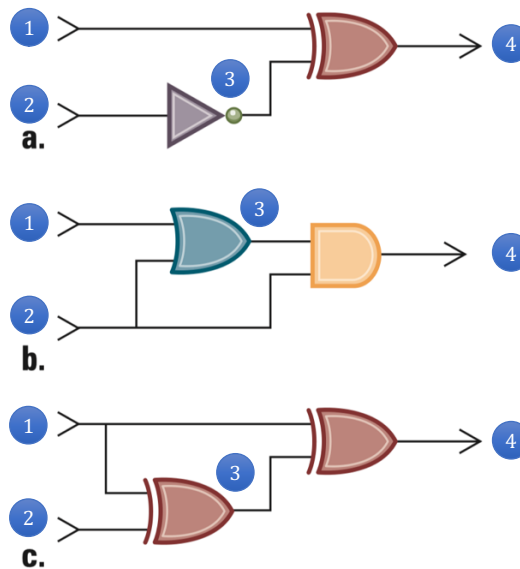
Duration: 120 minutes

- Each question is **4 points**, and the total point is **100 points**.
- Please write the solution **in detail** and **clearly** for each question; otherwise, you will lose partial point.

Q1. What is **the maximum number of entries** that must be interrogated when applying the **binary search** to a list of 200 entries? What about a list of 100,000 entries?

Ans: 8, 17

Q2. Determine the output of each of the following circuits, assuming that **the upper input is 1** and the **lower input is 0**. What would be the output when **the upper input is 0** and the **lower input is 1**?



Ans: a.

G1	G2	G3	G4
1	0	1	0
0	1	0	0

b.

G1	G2	G3	G4
1	0	1	0
0	1	1	1

c.

G1	G2	G3	G4
1	0	1	0
0	1	1	1

Q3. What does it mean to say that a programming language is **machine independent**?

Ans:

A programming language is said to be machine independent when the code written in this language **can be executed anywhere**.

Q4. What sequence of numbers would be printed by the following recursive function if we started it with N assigned the value 1?

```
def Exercise(N):  
    print(N)  
    if (N < 3):  
        Exercise(N + 1)  
    print(N)
```

Ans: 1, 2, 3, 3, 2, 1

Q5. What is the difference between an **interpreter** and a **compiler**?

Ans: The primary difference is in when the translation from high-level code to machine code happens:

- A Compiler translates the **entire program at once** before execution, creating a standalone executable file. This results in faster execution.
- An Interpreter translates and executes the code **line-by-line** during execution. This makes development faster (no compile step) but results in slower execution overall

Q6. Convert each of the following binary representations to its equivalent base ten form:

a. 10.11 b. 111.101 c. 1.01 d. 100.001 e. 101.1

Ans:

a. $2\frac{3}{4}$ b. $7\frac{5}{8}$ c. $1\frac{1}{4}$ d. $4\frac{1}{8}$ e. $5\frac{1}{2}$

Q7. What are the conditions required for **deadlock** in operating system?

Ans:

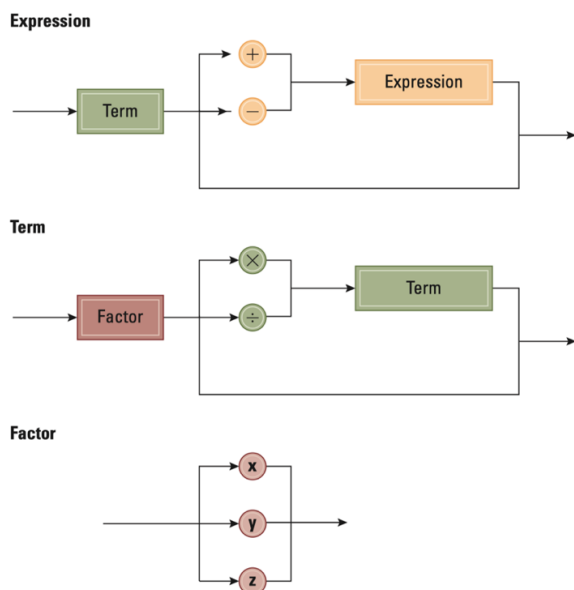
Conditions required for deadlock:

1. Competition for non-sharable resources

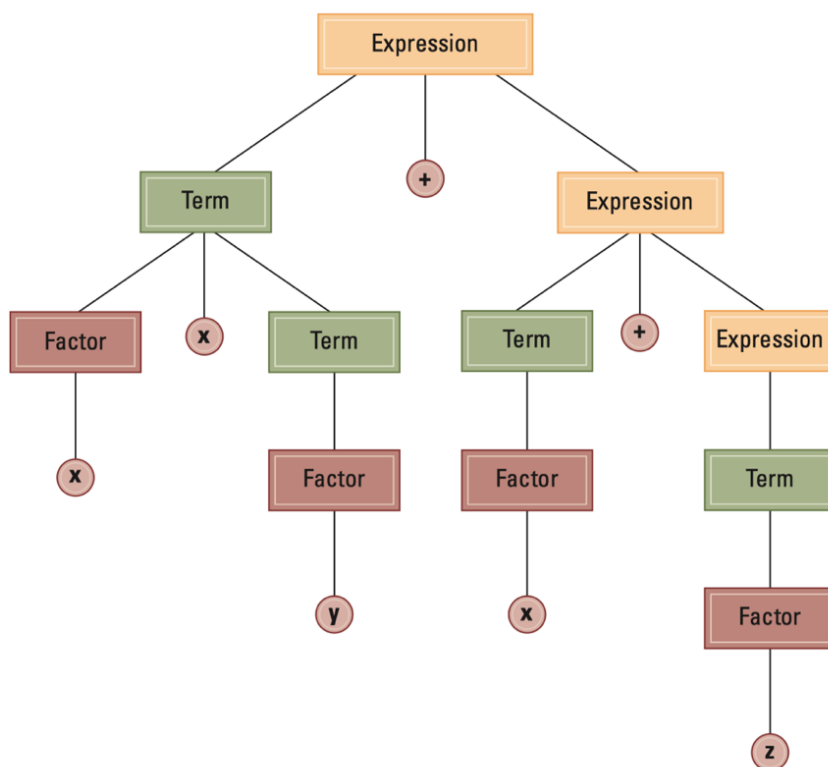
2. Resources requested on a partial basis

3. An allocated resource cannot be forcibly retrieved

Q8. Draw the parse tree for the expression $x \times y + y \div z$ based on the syntax diagrams below.



Ans:



Q9. What is the difference between **storage** and **main memory**?

Ans: The core difference between Main Memory and Storage is their purpose and volatility:

- **Main Memory (RAM):** This is the computer's temporary, fast-working space for data actively being used by the CPU. Its volatile data is lost when power is off. Capacity is smaller (e.g., 16GB).
- **Storage (SSD/HDD):** This is the computer's permanent, long-term repository for the OS, applications, and all files. Its non-volatile data persists without power. Capacity is much larger (e.g., 1TB).

Q10. Describe the three major steps in the **translation process**.

Ans:

Lexical analysis: the process of identifying tokens.

Parsing: the process of recognizing the grammatical structure of the program.

Code generation: the process of producing the instructions in the object program.

Q11. Which of the following activities require **real-time processing**?

- Playing League of Legends.
- Printing documents.
- Playing video from YouTube.
- Executing a program that predicts the weather today.
- Editing photos from Photoshop application.

Ans: a, c and e.

Q12. The following program segment is designed to calculate the factorial of a nonnegative number N using recursion. Is the program segment correct? Explain your answer. Modify this program segment to calculate the factorial without using recursion. Be sure that the output values do not change.

```
def fact (N):  
    if N == 0:  
        return 1;  
    return N x fact (N - 1);
```

Ans: Correct, but it can be better. Explain: $\text{fact}(7)=7*6*5*4*3*2*1*1 = 7!$

Factorial function without using recursion (sample code):

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```
def fact(n):  
    f = 1  
    while (n > 0):  
        f = f * n  
        n = n - 1  
    print(f)  
    return f
```

Q13. What is the difference between **application software** and **utility software**? Give 2 examples for each group, then explain why you classified them as you did.

Ans:

Application software: Performs specific tasks for users.

Examples:

- Chrome, Firefox, Edge: A browser, serve a website to client.
- LINE/Skype: A social network application which connect client to others.

System software: Provides infrastructure for application software.

Examples:

- Shell: Communicates with the machine's environment.
- File manager: Coordinates the use of the machine's mass storage.
- Device drivers: Handle communication with the machine's peripheral devices.
- Memory manager: Coordinates the use of the machine's main memory.
- Scheduler: Coordinates the processes in the system.
- Dispatcher: Controls the assignment of processes to CPU time.

Q14. What will be the values of a and b if we call the function `CodeWrite` with inputs 78 and 89?

```
def CodeWrite (a, b):  
    if (a > 0 and b > 0):  
        a = a + b;  
        b = a - b;  
        a = a - b;  
        print (a,b)
```

Ans: 89 and 78

```
CodeWrite (78, 89):  
if (a > 0 and b > 0):  
    167 = 78 + 89;  
    78 = 167 - 89;  
    89 = 167 - 78;  
    print(89 and 78)
```

Q15. What is the purpose of each of the following HTML tags?

- a. <html> b. c. </h1> d. </script>

Ans:

- a. <html> marks the beginning of an HTML document.
b. marks the beginning of a image.
c. </h1> marks the end of heading 1.
d. </script> marks the end of JavaScript section.

Q16. What are some differences between a transport layer based on the TCP protocol and another based on the UDP protocol?

Ans: Unlike TCP, UDP is a connectionless protocol that does not confirm that the message was received at the destination.

Q17. List the classes $\Theta(n^2)$, $\Theta(\log_2 n)$, $\Theta(n)$, and $\Theta(n^3)$ in **increasing order of efficiency**.

Ans: $\Theta(n^3)$, $\Theta(n^2)$, $\Theta(n)$ and $\Theta(\log_2 n)$.

Q18. Identify the components of each of the following abbreviated URLs.

- a. <https://www.google.com/index.html>
b. <https://www.cse.yzu.edu.tw/en/education/ungraduatecourse>
c. <https://portalx.yzu.edu.tw/PortalSocialVB/FMain/DefaultPage.aspx>

Ans:

- Protocol
- Mnemonic name/domain name
- Directory path
- Document name

Note: **ungraduatecourse** in this case also can be document name. Both are correct.

Q19. Summarize the distinction between a **grid computing** and **cloud computing**.

Ans:

- Grid computing focuses on high-performance computing (HPC) for one massive task using shared, distributed resources. **Each computer in grid computing store data independently.**
- Cloud computing focuses on on-demand service delivery (compute, storage, etc.) provided and managed by a vendor. **Cloud computing requires data centers for data storage.**

Q20. How many bytes of storage space would be required to store a 20-page document containing details of employees, in which each page contains 100 records and every record is of 200 characters, if **two-byte** Unicode characters were used?

Ans: $20 \times 100 \times 200 \times 2 = 800,000$ bytes

Q21. Give at least **5 examples** of insecure passwords and explain why?

Ans: 123456, abc123, use birthday or personal information as passwords. Students can provide more examples.

Q22. Which of the following is an example of an unreliable, connectionless protocol?

- a. UDP b. TCP c. SMTP d. FTP

Ans: UDP

Q23. A standard IPv6 address is how many bits long?

Ans: 128 bits

Q24. What protocol is used to retrieve email from a mail server, allowing the email to be managed on the server until the user deletes it?

- a. POP3 b. IMAP c. SMTP d. FTP

Ans: POP3

Q25. Using **Huffman Coding**, please show steps to encode/decode the 8 bits string **PROGRAMMING IS FUN FUN FUN** and calculate how many bits of data have been saved after encoding:

Ans: The full string length is 28 characters (including spaces).

char	quant
N	4
_ (space)	4
F	3
U	3
R	2
G	2
M	2
I	2
P	1
O	1
A	1
S	1

The final Huffman codes as follow:

char	code
N	00
_ (space)	101
F	011
U	010
R	1101
G	1100
M	1111
I	1110
P	10001
O	10000
A	10011
S	10010

Encoded String:

1000111011000011001101100111111111111000110010111101001010101101
0001010110100010101101000 (*90 bits*)

Bits Saved

Bits Saved = Original Bits–Huffman Bits

Bits Saved = (26 characters * 8 bits) – 90 bits = 118 bits

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Conclusion: Using Huffman Coding saved **118 bits of data**, achieving a compression ratio of approximately **43.269%**.